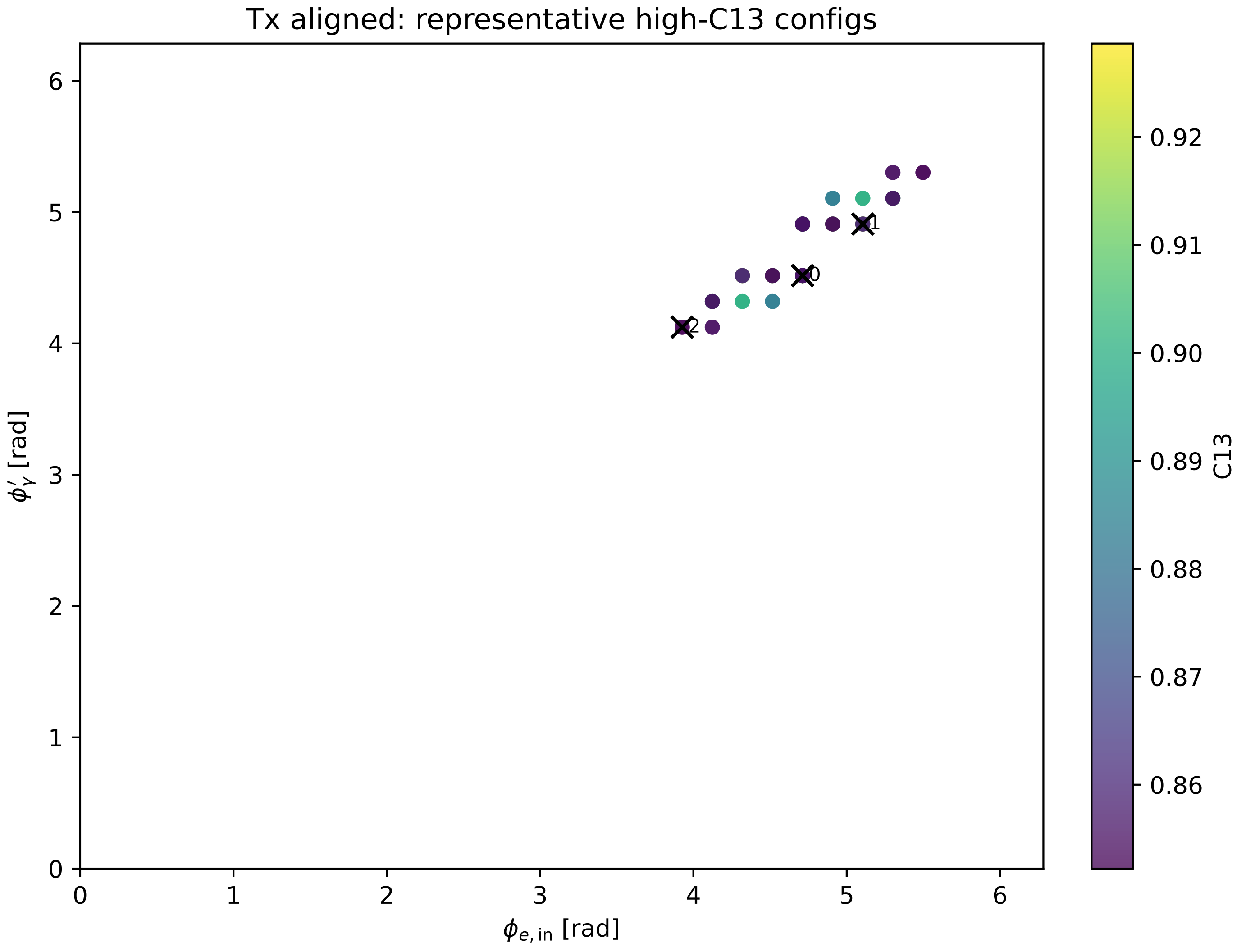
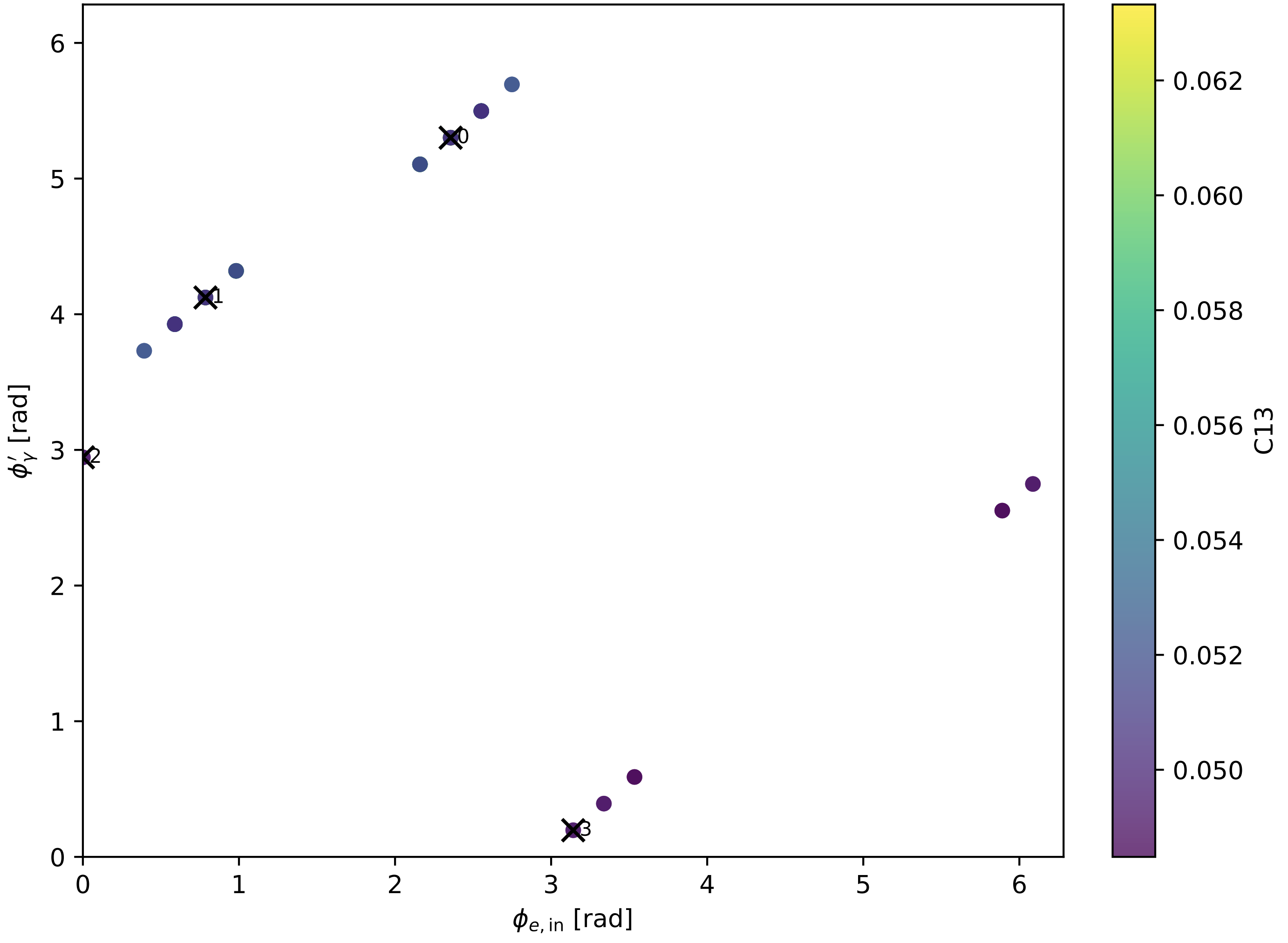


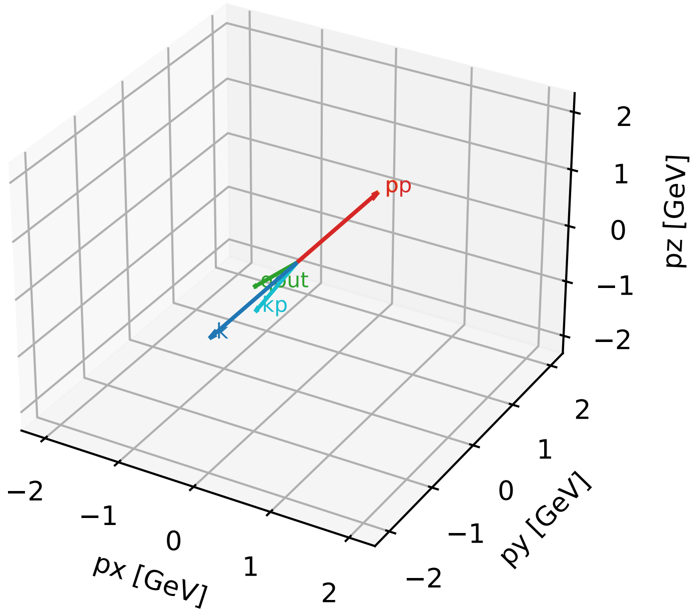
Tx aligned: representative high-C13 configs
ϕ'_γ [rad]
$\phi_{e,\mathrm{in}}$ [rad]
0
1
2
C13
0
1
2
3
4
5
6
0.86
0.87
0.88
0.89
0.90
0.91
0.92

Tx opposite: representative high-C13 configs



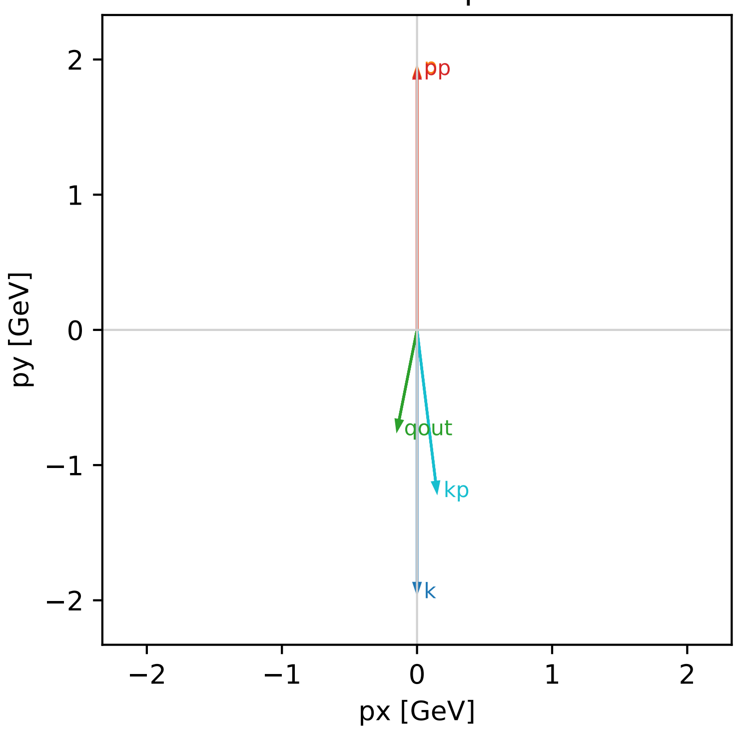
Tx characteristic high-C13 configuration

3D momenta

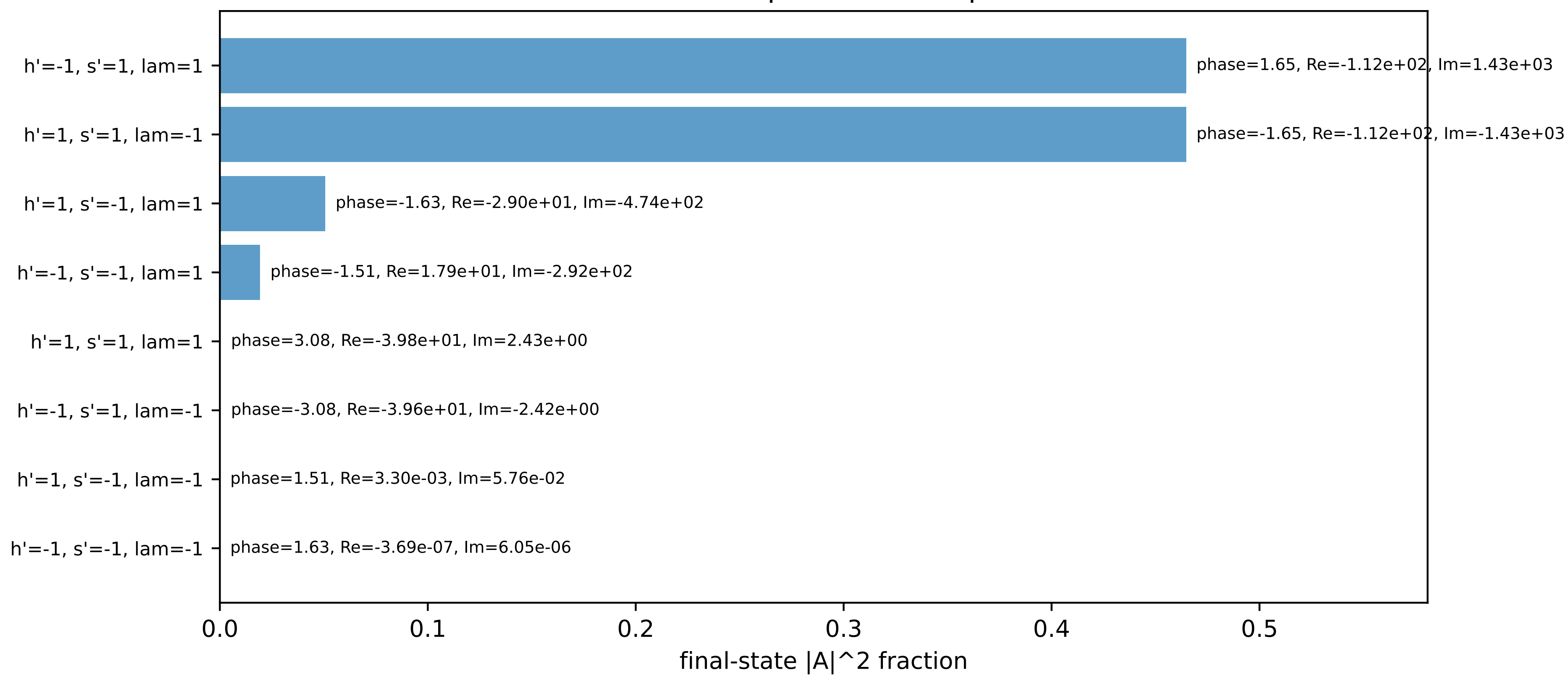


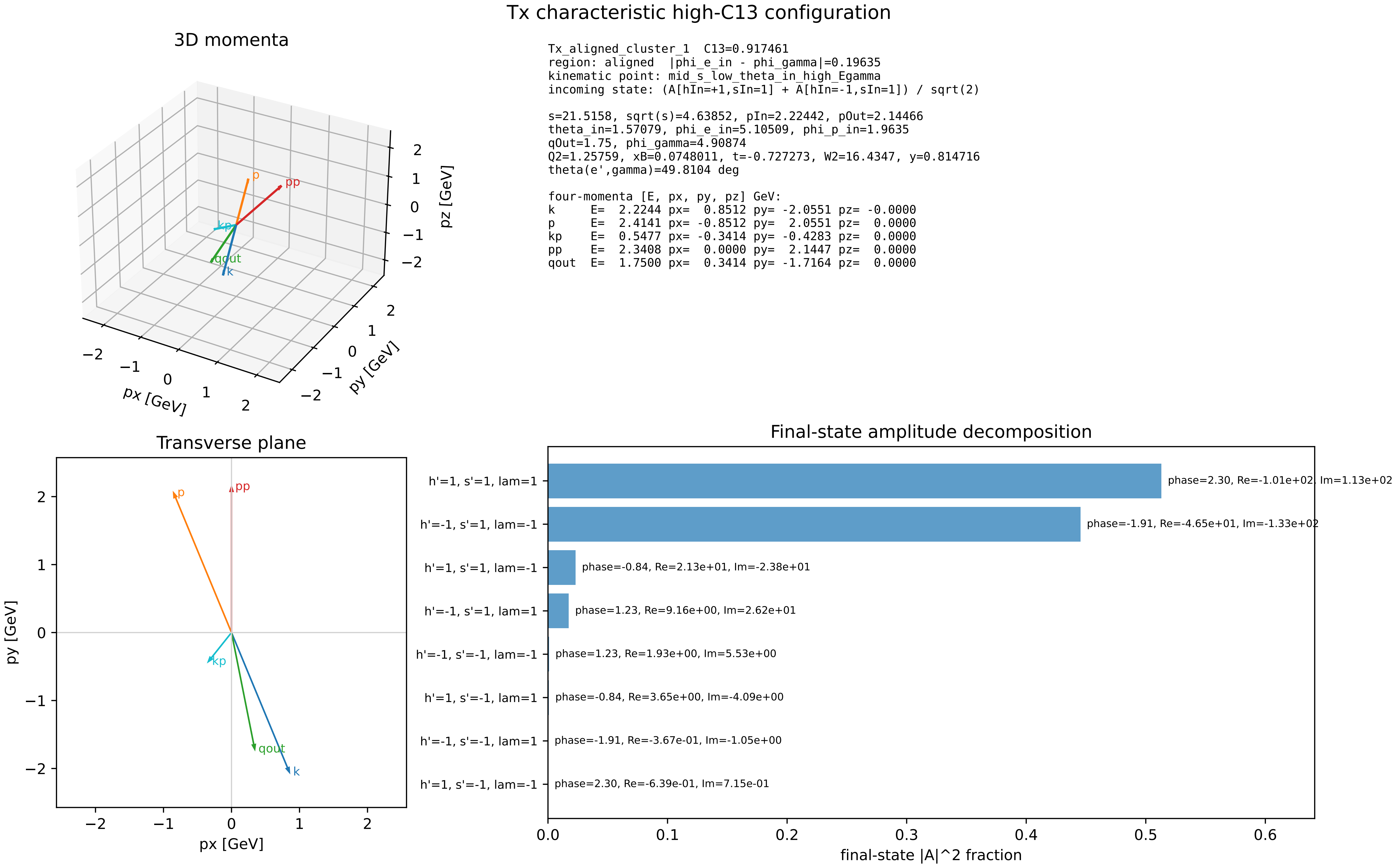

```
Tx_aligned_cluster_0  C13=0.928651
region: aligned  |phi_e_in - phi_gamma|=0.19635
kinematic point: low_s_low_theta_in_low_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)

s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=1.92864
theta_in=1.57079, phi_e_in=4.71239, phi_p_in=1.5708
qOut=0.75, phi_gamma=4.51604
Q2=0.0346994, xB=0.00569867, t=-2.87478e-05, W2=6.93417, y=0.382897
theta(e',gamma)=18.242 deg

four-momenta [E, px, py, pz] GeV:
k     E=  1.9409 px= -0.0000 py= -1.9409 pz= -0.0000
p     E=  2.1557 px=  0.0000 py=  1.9409 pz=  0.0000
kp    E=  1.2020 px=  0.1463 py= -1.1930 pz=  0.0000
pp    E=  2.1446 px=  0.0000 py=  1.9286 pz=  0.0000
qout  E=  0.7500 px= -0.1463 py= -0.7356 pz=  0.0000
```

Transverse plane

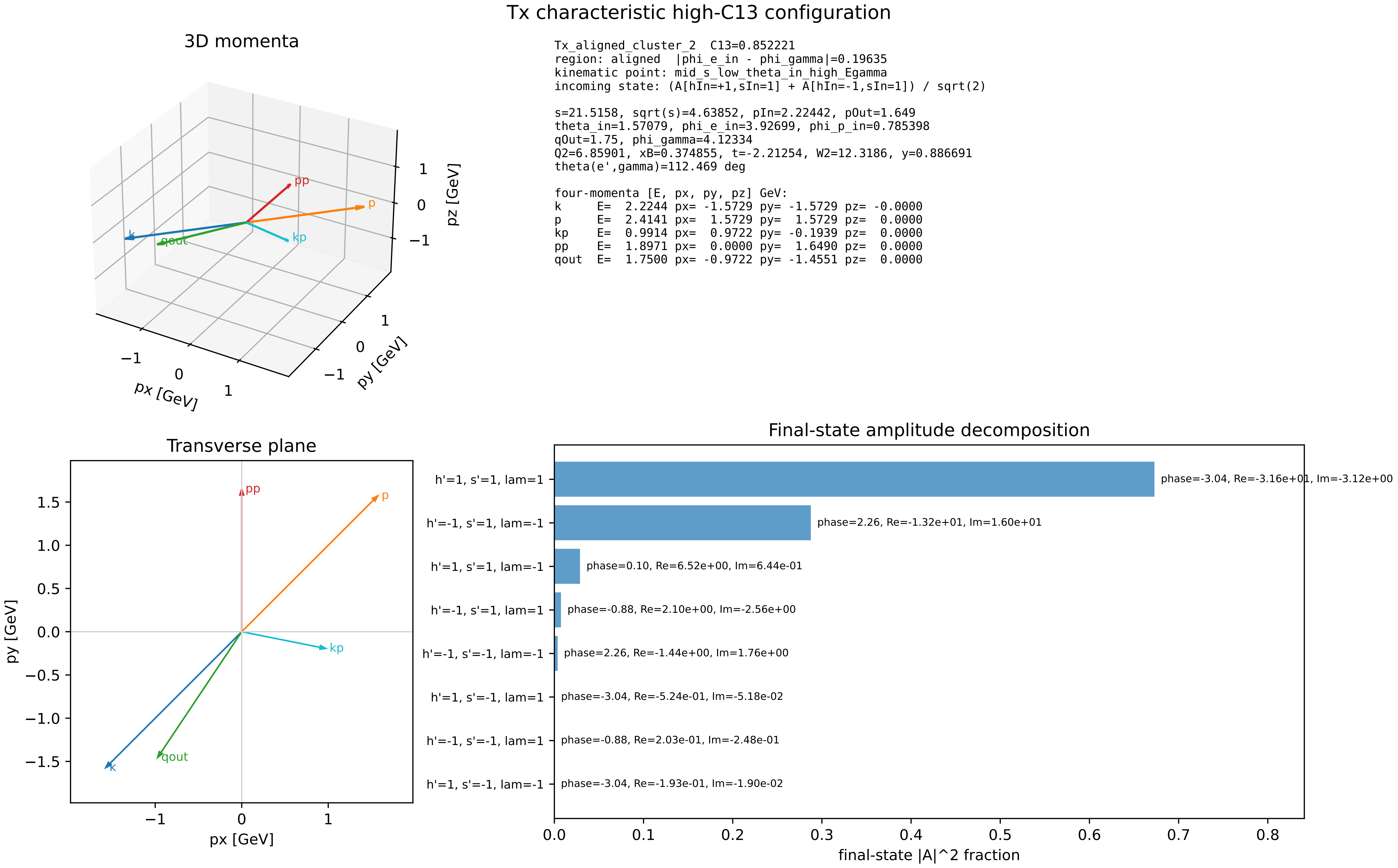


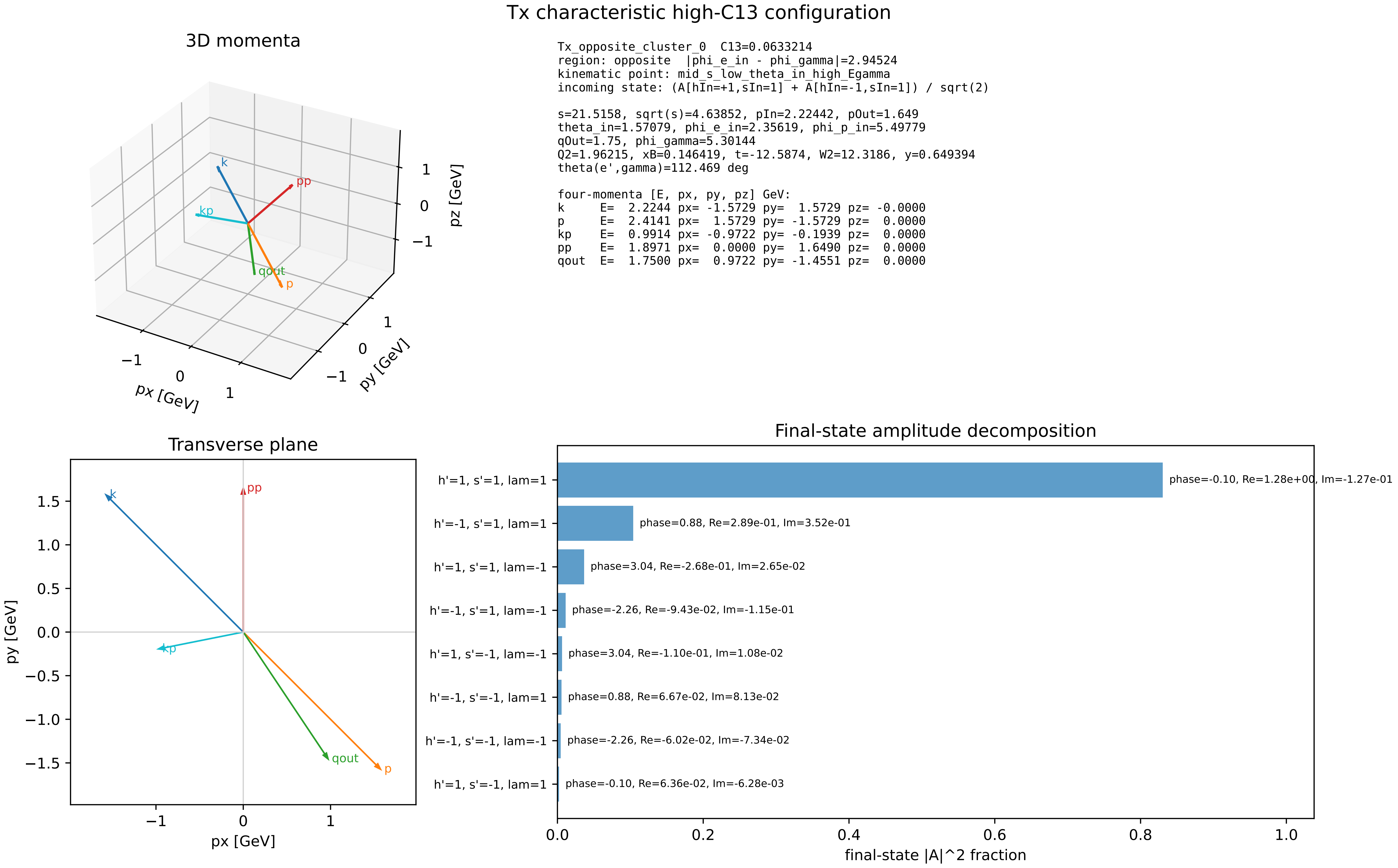
Final-state amplitude decomposition



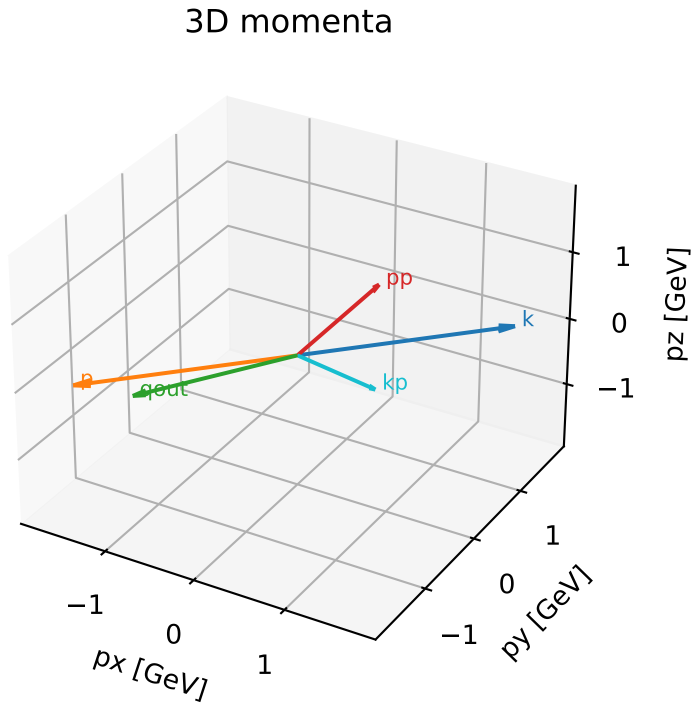


Tx characteristic high-C13 configuration
3D momenta
p
pp
kp
qout
k
px [GeV]
py [GeV]
pz [GeV]
Tx_aligned_cluster_1 C13=0.917461
region: aligned |phi_e_in - phi_gamma|=0.19635
kinematic point: mid_s_low_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)
s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=2.14466
theta_in=1.57079, phi_e_in=5.10509, phi_p_in=1.9635
qOut=1.75, phi_gamma=4.90874
Q2=1.25759, xB=0.0748011, t=-0.727273, W2=16.4347, y=0.814716
theta(e',gamma)=49.8104 deg
four-momenta [E, px, py, pz] GeV:
k   E= 2.2244 px= 0.8512 py= -2.0551 pz= -0.0000
p   E= 2.4141 px= -0.8512 py= 2.0551 pz= 0.0000
kp   E= 0.5477 px= -0.3414 py= -0.4283 pz= 0.0000
pp   E= 2.3408 px= 0.0000 py= 2.1447 pz= 0.0000
qout E= 1.7500 px= 0.3414 py= -1.7164 pz= 0.0000
Transverse plane
px [GeV]
py [GeV]
Final-state amplitude decomposition
h'=1, s'=1, lam=1 phase=2.30, Re=-1.01e+02, Im=1.13e+02
h'=-1, s'=1, lam=-1 phase=-1.91, Re=-4.65e+01, Im=-1.33e+02
h'=1, s'=1, lam=-1 phase=-0.84, Re=2.13e+01, Im=-2.38e+01
h'=-1, s'=1, lam=1 phase=1.23, Re=9.16e+00, Im=2.62e+01
h'=-1, s'=-1, lam=-1 phase=1.23, Re=1.93e+00, Im=5.53e+00
h'=1, s'=-1, lam=1 phase=-0.84, Re=3.65e+00, Im=-4.09e+00
h'=-1, s'=-1, lam=1 phase=-1.91, Re=-3.67e-01, Im=-1.05e+00
h'=1, s'=-1, lam=-1 phase=2.30, Re=-6.39e-01, Im=7.15e-01
final-state |A|^2 fraction

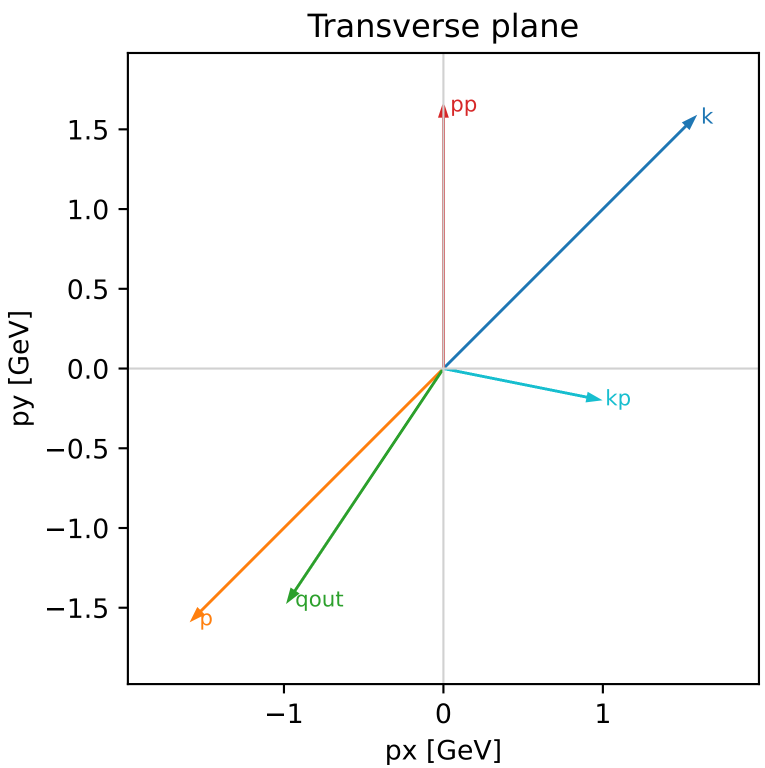

Tx characteristic high-C13 configuration
3D momenta
pp
p
k
qout
kp
px [GeV]
py [GeV]
pz [GeV]
Tx_aligned_cluster_2 C13=0.852221
region: aligned |phi_e_in - phi_gamma|=0.19635
kinematic point: mid_s_low_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)
s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=1.649
theta_in=1.57079, phi_e_in=3.92699, phi_p_in=0.785398
qOut=1.75, phi_gamma=4.12334
Q2=6.85901, xB=0.374855, t=-2.21254, W2=12.3186, y=0.886691
theta(e',gamma)=112.469 deg
four-momenta [E, px, py, pz] GeV:
k   E= 2.2244 px= -1.5729 py= -1.5729 pz= -0.0000
p   E= 2.4141 px= 1.5729 py= 1.5729 pz= 0.0000
kp   E= 0.9914 px= 0.9722 py= -0.1939 pz= 0.0000
pp   E= 1.8971 px= 0.0000 py= 1.6490 pz= 0.0000
qout E= 1.7500 px= -0.9722 py= -1.4551 pz= 0.0000
Transverse plane
px [GeV]
py [GeV]
Final-state amplitude decomposition
h'=1, s'=1, lam=1
phase=-3.04, Re=-3.16e+01, Im=-3.12e+00
h'=-1, s'=1, lam=-1
phase=2.26, Re=-1.32e+01, Im=1.60e+01
h'=1, s'=1, lam=-1
phase=0.10, Re=6.52e+00, Im=6.44e-01
h'=-1, s'=1, lam=1
phase=-0.88, Re=2.10e+00, Im=-2.56e+00
h'=-1, s'=-1, lam=-1
phase=2.26, Re=-1.44e+00, Im=1.76e+00
h'=1, s'=-1, lam=1
phase=-3.04, Re=-5.24e-01, Im=-5.18e-02
h'=-1, s'=-1, lam=1
phase=-0.88, Re=2.03e-01, Im=-2.48e-01
h'=1, s'=-1, lam=-1
phase=-3.04, Re=-1.93e-01, Im=-1.90e-02
final-state |A|^2 fraction

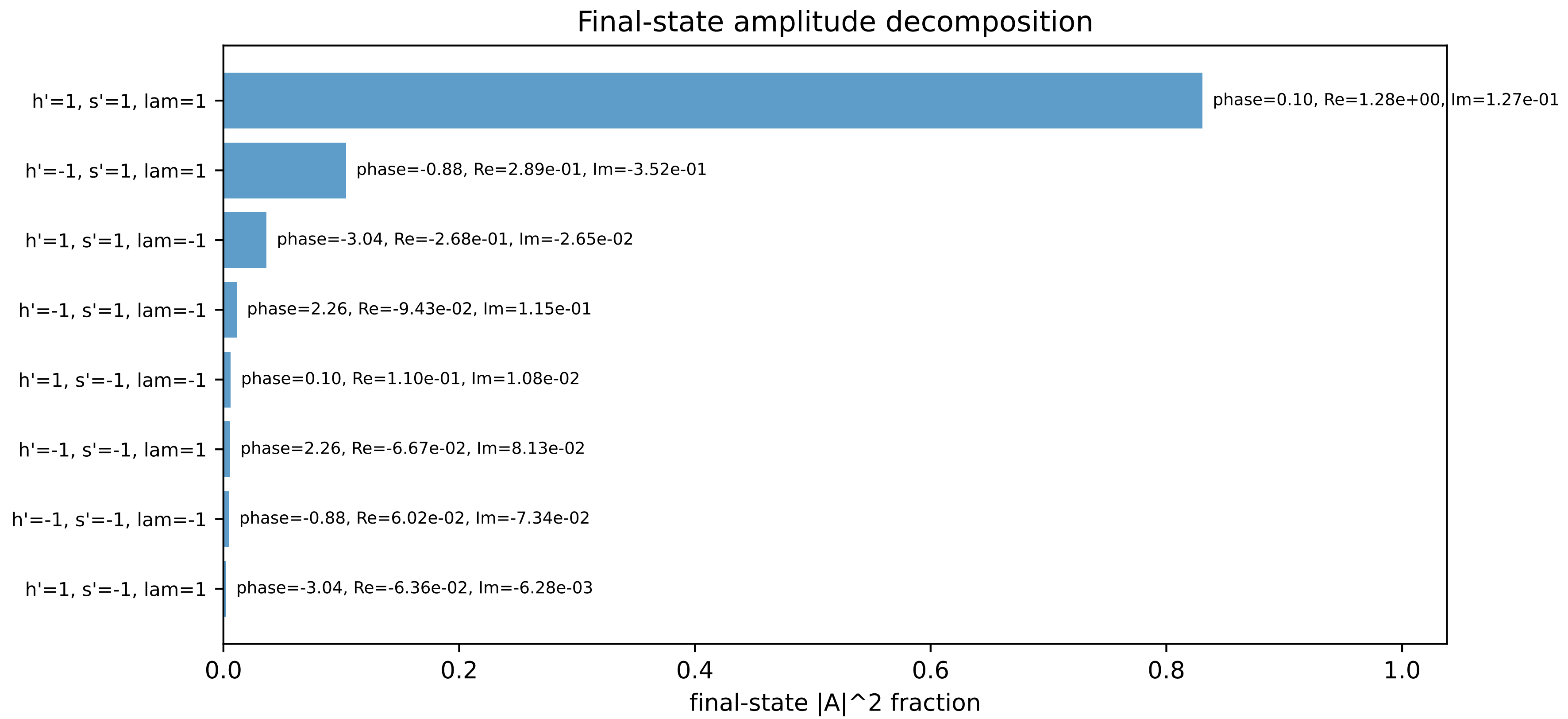

Tx characteristic high-C13 configuration
3D momenta
k
pp
kp
qout
p
px [GeV]
py [GeV]
pz [GeV]
Tx_opposite_cluster_0 C13=0.0633214
region: opposite |phi_e_in - phi_gamma|=2.94524
kinematic point: mid_s_low_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)
s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=1.649
theta_in=1.57079, phi_e_in=2.35619, phi_p_in=5.49779
qOut=1.75, phi_gamma=5.30144
Q2=1.96215, xB=0.146419, t=-12.5874, W2=12.3186, y=0.649394
theta(e',gamma)=112.469 deg
four-momenta [E, px, py, pz] GeV:
k   E= 2.2244 px= -1.5729 py= 1.5729 pz= -0.0000
p   E= 2.4141 px= 1.5729 py= -1.5729 pz= 0.0000
kp   E= 0.9914 px= -0.9722 py= -0.1939 pz= 0.0000
pp   E= 1.8971 px= 0.0000 py= 1.6490 pz= 0.0000
qout E= 1.7500 px= 0.9722 py= -1.4551 pz= 0.0000
Transverse plane
px [GeV]
py [GeV]
Final-state amplitude decomposition
h'=1, s'=1, lam=1
phase=-0.10, Re=1.28e+00, Im=-1.27e-01
h'=-1, s'=1, lam=1
phase=0.88, Re=2.89e-01, Im=3.52e-01
h'=1, s'=1, lam=-1
phase=3.04, Re=-2.68e-01, Im=2.65e-02
h'=-1, s'=1, lam=-1
phase=-2.26, Re=-9.43e-02, Im=-1.15e-01
h'=1, s'=-1, lam=-1
phase=3.04, Re=-1.10e-01, Im=1.08e-02
h'=-1, s'=-1, lam=1
phase=0.88, Re=6.67e-02, Im=8.13e-02
h'=-1, s'=-1, lam=-1
phase=-2.26, Re=-6.02e-02, Im=-7.34e-02
h'=1, s'=-1, lam=1
phase=-0.10, Re=6.36e-02, Im=-6.28e-03
final-state |A|^2 fraction

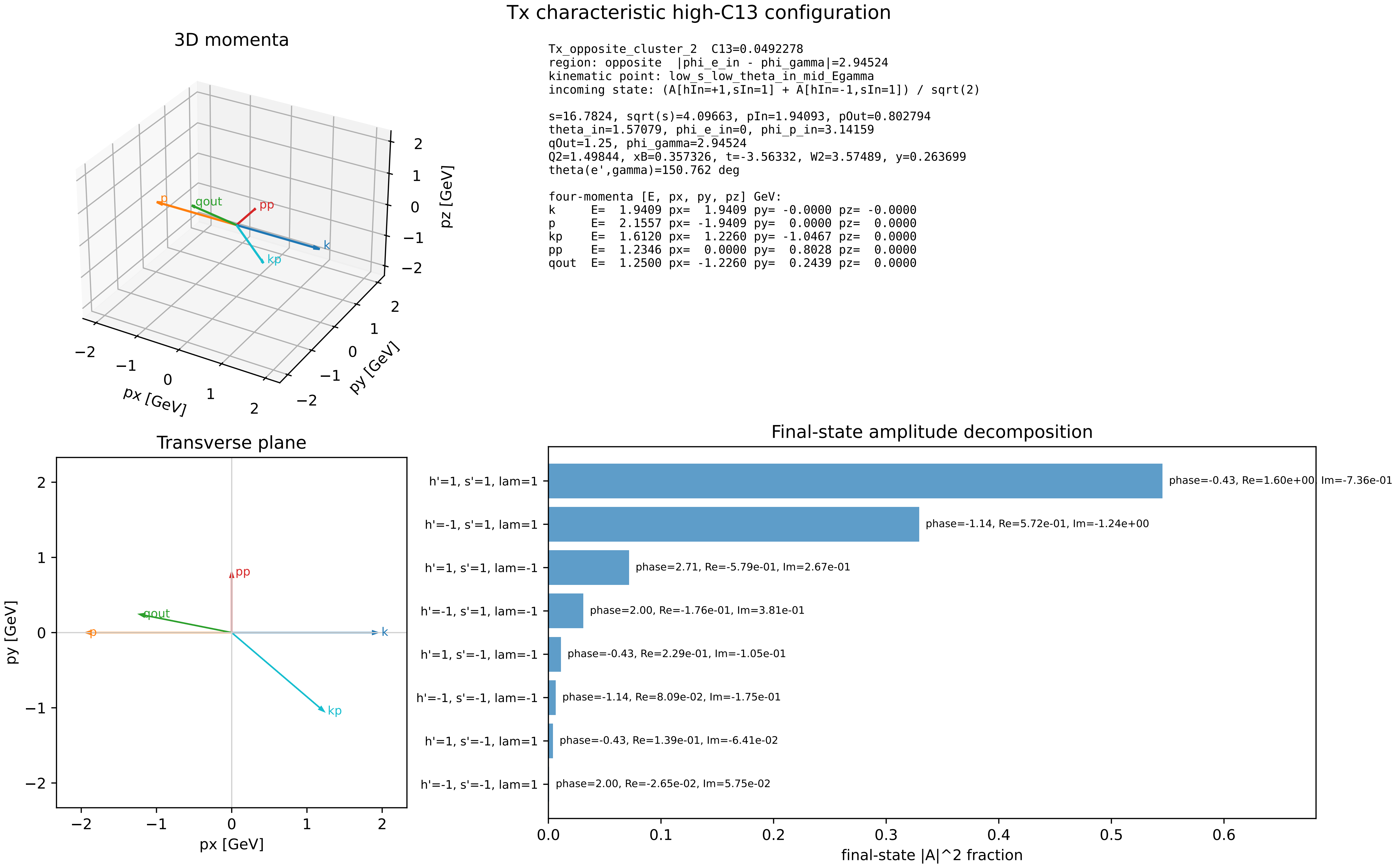
Tx characteristic high-C13 configuration



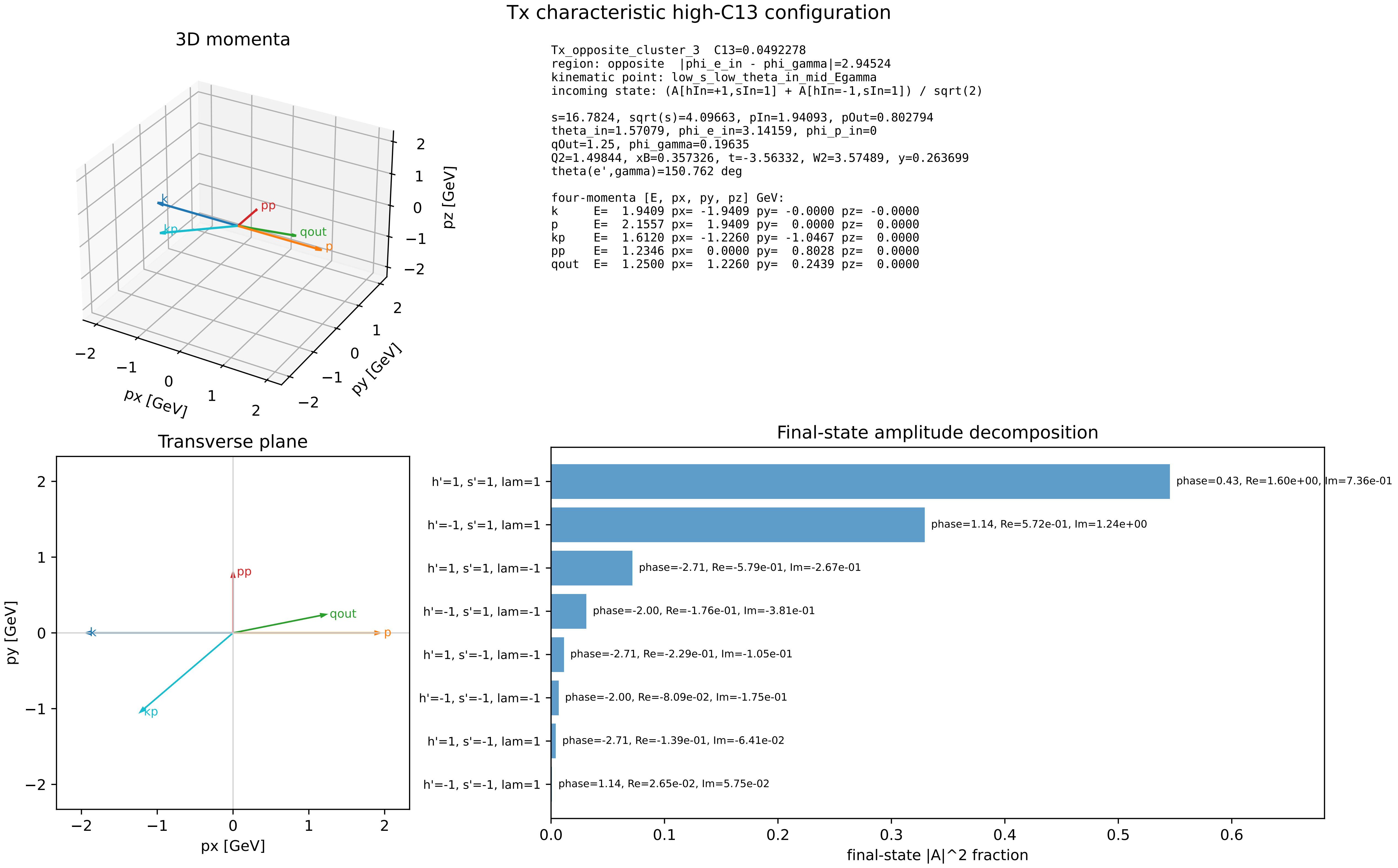

```
Tx_opposite_cluster_1  C13=0.0633214
region: opposite  |phi_e_in - phi_gamma|=2.94524
kinematic point: mid_s_low_theta_in_high_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)

s=21.5158, sqrt(s)=4.63852, pIn=2.22442, pOut=1.649
theta_in=1.57079, phi_e_in=0.785398, phi_p_in=3.92699
qOut=1.75, phi_gamma=4.12334
Q2=1.96215, xB=0.146419, t=-12.5874, W2=12.3186, y=0.649394
theta(e',gamma)=112.469 deg

four-momenta [E, px, py, pz] GeV:
k     E=  2.2244 px=  1.5729 py=  1.5729 pz= -0.0000
p     E=  2.4141 px= -1.5729 py= -1.5729 pz=  0.0000
kp    E=  0.9914 px=  0.9722 py= -0.1939 pz=  0.0000
pp    E=  1.8971 px=  0.0000 py=  1.6490 pz=  0.0000
qout  E=  1.7500 px= -0.9722 py= -1.4551 pz=  0.0000
```







Tx characteristic high-C13 configuration
3D momenta
p
qout
pp
k
kp
px [GeV]
py [GeV]
pz [GeV]
Tx_opposite_cluster_2 C13=0.0492278
region: opposite |phi_e_in - phi_gamma|=2.94524
kinematic point: low_s_low_theta_in_mid_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)
s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=0.802794
theta_in=1.57079, phi_e_in=0, phi_p_in=3.14159
qOut=1.25, phi_gamma=2.94524
Q2=1.49844, xB=0.357326, t=-3.56332, W2=3.57489, y=0.263699
theta(e',gamma)=150.762 deg
four-momenta [E, px, py, pz] GeV:
k   E= 1.9409 px= 1.9409 py= -0.0000 pz= -0.0000
p   E= 2.1557 px= -1.9409 py= 0.0000 pz= 0.0000
kp   E= 1.6120 px= 1.2260 py= -1.0467 pz= 0.0000
pp   E= 1.2346 px= 0.0000 py= 0.8028 pz= 0.0000
qout E= 1.2500 px= -1.2260 py= 0.2439 pz= 0.0000
Transverse plane
px [GeV]
py [GeV]
Final-state amplitude decomposition
h'=1, s'=1, lam=1
phase=-0.43, Re=1.60e+00, Im=-7.36e-01
h'=-1, s'=1, lam=1
phase=-1.14, Re=5.72e-01, Im=-1.24e+00
h'=1, s'=1, lam=-1
phase=2.71, Re=-5.79e-01, Im=2.67e-01
h'=-1, s'=1, lam=-1
phase=2.00, Re=-1.76e-01, Im=3.81e-01
h'=1, s'=-1, lam=-1
phase=-0.43, Re=2.29e-01, Im=-1.05e-01
h'=-1, s'=-1, lam=-1
phase=-1.14, Re=8.09e-02, Im=-1.75e-01
h'=1, s'=-1, lam=1
phase=-0.43, Re=1.39e-01, Im=-6.41e-02
h'=-1, s'=-1, lam=1
phase=2.00, Re=-2.65e-02, Im=5.75e-02
final-state |A|^2 fraction


Tx characteristic high-C13 configuration
3D momenta
Tx_opposite_cluster_3 C13=0.0492278
region: opposite |phi_e_in - phi_gamma|=2.94524
kinematic point: low_s_low_theta_in_mid_Egamma
incoming state: (A[hIn=+1,sIn=1] + A[hIn=-1,sIn=1]) / sqrt(2)
s=16.7824, sqrt(s)=4.09663, pIn=1.94093, pOut=0.802794
theta_in=1.57079, phi_e_in=3.14159, phi_p_in=0
qOut=1.25, phi_gamma=0.19635
Q2=1.49844, xB=0.357326, t=-3.56332, W2=3.57489, y=0.263699
theta(e',gamma)=150.762 deg
four-momenta [E, px, py, pz] GeV:
k   E= 1.9409 px= -1.9409 py= -0.0000 pz= -0.0000
p   E= 2.1557 px= 1.9409 py= 0.0000 pz= 0.0000
kp   E= 1.6120 px= -1.2260 py= -1.0467 pz= 0.0000
pp   E= 1.2346 px= 0.0000 py= 0.8028 pz= 0.0000
qout E= 1.2500 px= 1.2260 py= 0.2439 pz= 0.0000
k
pp
kp
qout
p
pz [GeV]
px [GeV]
py [GeV]
Transverse plane
py [GeV]
px [GeV]
Final-state amplitude decomposition
h'=1, s'=1, lam=1
phase=0.43, Re=1.60e+00, Im=7.36e-01
h'=-1, s'=1, lam=1
phase=1.14, Re=5.72e-01, Im=1.24e+00
h'=1, s'=1, lam=-1
phase=-2.71, Re=-5.79e-01, Im=-2.67e-01
h'=-1, s'=1, lam=-1
phase=-2.00, Re=-1.76e-01, Im=-3.81e-01
h'=1, s'=-1, lam=-1
phase=-2.71, Re=-2.29e-01, Im=-1.05e-01
h'=-1, s'=-1, lam=-1
phase=-2.00, Re=-8.09e-02, Im=-1.75e-01
h'=1, s'=-1, lam=1
phase=-2.71, Re=-1.39e-01, Im=-6.41e-02
h'=-1, s'=-1, lam=1
phase=1.14, Re=2.65e-02, Im=5.75e-02
final-state |A|^2 fraction